Cognitive Mapping and Structural Transfer Across Environments

Yes, in neuroscience and cognitive psychology, this phenomenon exists and is extensively studied. It is primarily described through two complementary concepts: **analogical mapping** at the psychological level, and the **transfer of structural knowledge (or cognitive maps reuse)** at the neuroscientific level.

This phenomenon relies on the human brain's ability to extract the **relational structure** of one environment (how objects interact with each other) and project it onto a completely different environment to understand or navigate it.

Here is how it is defined and explained across disciplines:

1. Cognitive Psychology: Analogical Mapping and Structural Alignment

According to the *structure-mapping theory* (developed by psychologist Dedre Gentner), humans do not just compare objects based on their visual appearance; instead, they compare the **relations between objects**.

- **Analogical Mapping:** The exact cognitive process where a familiar environment (called the *Base* or *Source*) serves as a template to comprehend a new environment (the *Target*). The brain maps a one-to-one correspondence not between the objects themselves, but between their interactions.
- **Classic Example:** When learning to navigate a website (virtual environment) by applying the interaction rules of a physical office (e.g., a "folder" containing "files", or a "trash bin" to discard items). Another example is understanding the atom (microscopic environment) by mapping the structural interactions of the solar system (macroscopic environment).

2. Neuroscience: Generalization and Cognitive Map Transfer

At the brain level, neuroscience explains this phenomenon through the structures of the **medial temporal lobe**, specifically the **hippocampus** and the **entorhinal cortex**.

- **Abstract Cognitive Maps:** Originally discovered for physical space (credited to *place cells* and *grid cells*), we now know the brain uses the same geometric and spatial code to map abstract concepts, social hierarchies, and relational interactions.
- **Transfer of Structural Knowledge:** Recent neuroscientific research demonstrates that once the brain learns the "interaction rule" (the topology) in one environment, the

hippocampus *generalizes* this structure. When entering a new environment with similar rules, the brain does not build a map from scratch: it **reuses the same relational grid** (the same neural activation pattern) and simply binds the newly perceived objects to it.

- **Predictive Coding:** The brain uses the map from the first environment to generate predictions about how objects will move or interact in the second environment, significantly accelerating cognitive processing times.

Visual Summary of the Underlying Mechanisms

Specific Phenomenon	Discipline	What the Brain Does
Structural Alignment	Cognitive Psychology	Isolates the interaction rules between objects A and B in Environment 1 and looks for their equivalents in Environment 2.
Map Induction	Computational Neuroscience	Assembles fragments of past experiences to predict the layout and rules of a partially or completely unexplored environment.
Neural Reuse	Neuroscience	Repurposes the exact neural circuits evolved for physical action or navigation in one environment to handle conceptual and relational tasks in another.